Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Meck, Janice V Ph.D.		
Project Title:	Presidential Award: Characterization of Autonomic Function, Sodium Handling, and Nitric Oxide Physiology in Astronauts and Patients Known to be Susceptible to Orthostatic Intolerance		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	09/30/2003	**End Date:**	10/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The Presidents Award is used to support the experiment, "Characterization of Autonomic Function, Sodium Handling, and Nitric Oxide Physiology in Astronauts and Patients Known to be Susceptible to Orthostatic Intolerance". Astronauts demonstrate varying degrees of autonomic dysfunction and orthostatic intolerance after spaceflight. Despite inflight and postflight countermeasure regimens, there remains a subset of astronauts that become presyncopal while standing on landing day. The studies in this proposal are unified by the overriding hypothesis that the susceptibility of some, but not all, astronauts to post-spaceflight orthostatic hypotension results from a predisposition in a subset of the normal population who are only somewhat more intolerant to orthostatic stress under normal conditions, but are pushed to frank orthostatic intolerance by spaceflight. This study addresses this hypothesis by performing a comprehensive set of studies to further characterize the population of astronauts known to be susceptible to post-spaceflight orthostatic intolerance. The purpose of this study is to characterize autonomic cardiovascular regulation, fluid volume and sodium handling, and		

	nitric oxide physiology in astronauts known to be susceptible to post-spaceflight orthostatic intolerance. This sub-set of astronauts will be compared to a patient population which suffers clinical symptoms of orthostatic intolerance due to autonomic dysfunction, a population of "false positive" subjects who have no clinical symptoms but cannot tolerate upright tilt for extended periods of time, a group of non-susceptible astronauts, and a control group of subjects from the general population.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This funding will be used in FY04 and FY05 to study orthostatic hypotension in patient populations with autonomic dysfunction.
Task Progress:	We have completed testing in 25 of 25 subjects (9 astronauts and 16 non-astronaut subjects). We are in the process of recruiting two more female subjects to complete the study. All data analysis is current with the exception of the hormone analysis.
Bibliography Type:	Description: (Last Updated: 04/15/2009)